

Space age

We've been to Mars – our space probes and robots have been there – but are there more trips lined up?

**Fis(8)ke**

Fiske Planetarium in Colorado recently reopened with newly incorporated 8K screens <http://dgit.in/fis8kay>

AMD GPU14 TECH DAY: NEW BEGINNINGS

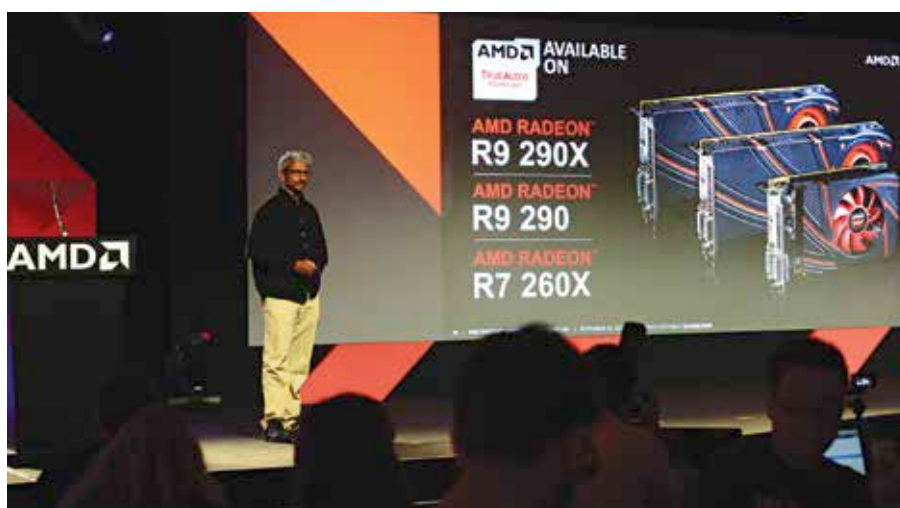
There's a Radeon revival on the horizon with new GPUs, new features, backed by a bold new AMD. Excited?

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At the beginning of the year, AMD showcased the new Radeon 8000 series GPUs at CES 2013, on a few partner products (all either laptops or ultrabooks), all made from the existing chips found on the Radeon HD 7000 series cards. As the world waited frustratingly for a tangible graphics update on AMD's Radeon desktop lineup, it never materialized around their speculated launch dates. A lot of gamers blamed Grinch, but he wasn't really to be blamed for not delivering shiny new Radeon GPUs to graphics-starved PC gamers.

While we were all contemplating whether AMD had overshot another one of its deadlines – and the chip maker's rather infamous of doing it – something big was brewing within the company. We got an early sign of things to come during the middle-of-the-year next-gen gaming console sneak peek (remember the PS4 and Xbox One announcement events?). Was it all a big coincidence that AMD managed to convince not just Microsoft but also Sony to become the hardware platform of choice for the next wave of gaming consoles (including the Nintendo Wii U released late last year)? In one fell swoop, AMD positioned itself at the centre of traditional console gaming, and we wondered what more the company hid



Raja Koduri speaking about AMD's new GPUs, and introducing Mantle – a new GPU API

behind closed doors. Tech and products that would still have to wait before they were revealed to inquisitive, rejoicing, and uncompromising eyes.

However, in Hawaii of all places, AMD finally rolled back the curtains and gave the world a glimpse of what it's been working on for the past several months.

AMD Radeon R9 and R7 series GPUs

For those of us who wondered when AMD was going to finally launch its 8000 series of graphics cards, well there's some news for you. There will be no 8000 series of cards, as AMD's changing the nomenclature for its GPUs. That's right, the four digit number naming convention wasn't sustainable "with an eye on the future," according to AMD, and let's face it the "HD" tag in the GPU names was getting old, too. They not only revealed their new GPU naming convention, but AMD also unveiled the much-anticipated successor to the HD 7990, the Radeon R9-290X – AMD's

fastest graphics card yet – and Radeon R9-290, both of which sport a brand new silicon architecture inside. There's also the R9 280X and R9 270X which when coupled with the previous two cards complete the top-end, aimed at uber gamers and priced highest accordingly. At the other end of the spectrum lies AMD Radeon R7 series comprising Radeon R7 240, R7 250, and R7 260X, which is targeted at budget to average gamers who desire more value out of their buck.

The AMD Radeon R9-290X is the flagship GPU of AMD's 2014 lineup, and it's a beast of a device. However, the entire Radeon R9 290 series sports some common features like over 5 TFLOPS of computing horsepower, over 300 GBps of memory bandwidth, ability to process 4 billion triangles per second, and they have more than 6 billion individual transistors. Another peculiar feature of all the new AMD Radeon R9 and R7 series of GPUs is that none of them have external CrossFire – neither the ports or the bridge. All of that's being taken care of through



New AMD R9 and R7 series GPUs are out!